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Vanadium and iron substituted polyoxotungstate iodonium and tetrabutylammonium salts for visible light free radical polymerization

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ABSTRACT

A series of diphenyliodonium and tetrabutylammonium salts of vanadium (+V) and iron (+III) mono and disubstituted polyoxotungstates was designed and tested as visible light photoinitiators for the free radical polymerization of acrylates or the cationic polymerization of epoxides. Interestingly, these polyoxotungstates could act as efficient photosensitizers for Type I or Type II photoinitiators (TPO-L, ITX), enabling to get excellent monomer conversions upon excitation at 405 nm.

1. Introduction

During the past decades, photopolymerization has been the focus of intense research efforts to face the different challenges visible light photopolymerization has to overcome. If UV photopolymerization has been extensively used since the 60 s, the safety concerns related to UV light [1–5] but also the energy-intensive nature of this technology has discarded this historical technology in favor of a more energy-efficient approach *i.e.* visible light photopolymerization [6–11]. The development of energy-saving, cheap, compact, long-living and lightweight devices that are light-emitting diodes (LEDs) has also greatly contributed to the emergence of visible light photopolymerization. Parallel to this, a higher light penetration within the resins can be obtained in the visible range, enabling to polymerize polymer films thicker than what can be achieved under UV light [12]. Benefiting from advantages such as a temporal and spatial control, photopolymerization is nowadays used in a wide range of applications going from coatings, dentistry, microelectronics, 3D printing or adhesives [13–21]. With the aim at identifying new families of photoinitiators exhibiting a remarkable reactivity under visible light, polyoxometalates (POMs) have been identified as promising candidates for this purpose. Polyoxometalates are also nowadays used photocatalysts and electrocatalysts for numerous chemical reactions, justifying their investigations as photocatalysts for polymer chemistry [22]. One of the first report mentioning the use of POMs in photopolymerization was published in 2011 by Lalevée and coworkers [23]. In this pioneering work, decatungstate $[W_{10}O_{32}]^{4-}$ which absorbs in the 300–420 nm range could efficiently promote the ring-opening of epoxides using the three-component $[W_{10}O_{32}]^{4-}$/diphenyliodonium salt/*tris*(trimethylsilyl)silane system upon irradiation with a Xenon lamp. In 2013, the possibility to generate interpenetrated polymer network (IPN) by the concomitant polymerization of acrylates and epoxides was demonstrated with the three-component $H_4[SiMo_{12}O_{40}]$/diphenyliodonium salt/*tris*(trimethylsilyl)silane system upon excitation with a Xe-Hg lamp [24]. In 2015, a step further was achieved by developing monocomponent POM-iodonium salts both with $[W_{10}O_{32}]^{4-}$ and $[SiMo_{12}O_{40}]^{4-}$ [25]. In these salts, the anionic photosensitizer and the cationic co-initiator could form a salt, simplifying the composition of the photocurable resin. In 2019, an elegant strategy consisted in using the POMs $[PMo_{12}O_{40}]^{3-}$ and $[SiMo_{12}O_{40}]^{4-}$ as photoinitiators of polymerization and subsequent to the polymerization process, to use the resulting polymer films for the photodegradation of pollutants. Thanks to a photocatalytic mechanism, the chemical

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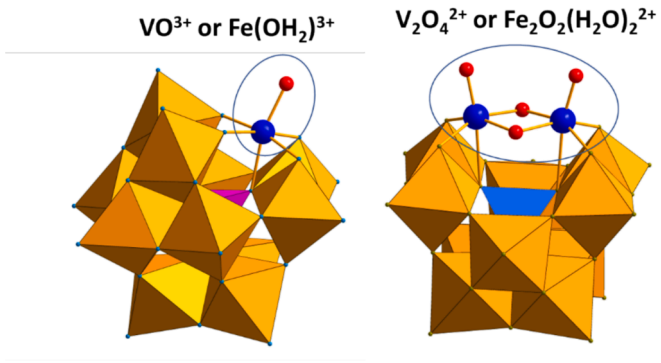

Fig. 1. Representation of the POMs used in this study. Orange polyhedra correspond to "WO_6" octahedra, while "PO_4" and "SiO_4" tetrahedra are depicted in pink and blue respectively.

integrity of POMs could be maintained during the polymerization process, enabling in a second step to use these POMs as photocatalysts for the degradation of pollutants. The photodegradation of numerous dyes such as eosin-Y,[26] erythrosine and rose Bengal, [27,28] bisphenol-A, [29] but also of common pharmaceutical compounds such as ciprofloxacin, oxytetracycline, ibuprofen or erythromycin could be successfully achieved [30,31]. In 2019 and in 2023, photochromic materials were prepared using derivatives of Anderson type $[MnMo_6O_{18}(Tris)_2]^{3-}$ and Lindqvist type $[Mo_6O_{19}]^{2-}$ POMs substituted either with a negative organic photochrom, namely a donor–acceptor Stenhouse adduct (DASA) [32] or with anthracene, anthraquinone or benzophenone groups [33]. In both cases the POM-photochrom conjugates can be used as photoinitiators for the free radical polymerization of polyethylene glycol diacrylate (PEG-DA) or to promote free-radical photopolymerization of acrylate monomer derivatives under irradiation in the visible range, respectively.

The main part of polyoxometalates investigated as visible light photoinitiators of polymerization are polyoxomolybdates. These POMs usually display strong absorption bands due to the ligand to metal charge transfer (LMCT) band between oxo ligands and Mo(VI) centers in the UV region with a significant part found in the visible range, and reduced gap value, which render them interesting for photochemical reactions after irradiation in UV–Visible domain. From their side, polyoxotungstates are much less used for photochemistry, excepted for decatungstate $[W_{10}O_{32}]^{4-}$ [34]. Nonetheless, it is well-known that polyoxotungstates are more robust than polyoxomolybdates,[35] an advantage for applications but they mainly absorb UV-light, while an excitation in the visible range is required for applications.

To the best of our knowledge, transition metal substituted polyoxotungstates have not been proposed yet. Mono or dilacunary Keggin-type polyoxotungstates, such as $[PW_{11}O_{39}]^{7-}$ and $[\gamma\text{-}SiW_{10}O_{38}]^{8-}$, substituted by Fe(III) or V(V) based clusters exhibit several interesting features compared to usual Keggin polyoxomolybdates and polyoxotungstates. First, these POMs are colored and thus can exhibit an enhanced absorption in the visible range. If these transitions are efficient to promote excited states of the POM moiety, this is particularly advantageous for photopolymerization applications. Parallel to this, an efficient absorption of light in the visible range also means that less energetic light sources, such as LEDs, can be used to initiate the polymerization process, what perfectly fit with the present requirement for developing photoinitiating systems activable under low light intensity *i.e.* in energy saving conditions. The introduction of Fe(III) and V(V) atoms in POMs also bring interesting properties in terms of redox and catalytic properties [36].

Photopolymerization is used in a wide range of applications, and notably for 3D printing. Among polymerization reactions, photopolymerization is mostly applied to the free radical polymerization (FRP) of (meth) acrylates and the cationic polymerization (CP) of epoxides. From the mechanistic viewpoint, POMs being electron-rich species, a photoinduced electron transfer towards an electrodeficient iodonium salt (Ar_2I^+) can be easily obtained, as shown in equations r1 and r2. Subsequent to the electron transfer, a decomposition of the unstable $Ar_2I^\bullet$ radical occurs, furnishing the highly reactive $Ar^\bullet$ aryl radical (see equation r3). The FRP of (meth)acrylates can thus be initiated [24].

$$H_4[SiMo_{12}O_{40}] \rightarrow {}^*H_4[SiMo_{12}O_{40}]\ (h\nu)\ (r1)$$

$${}^*H_4[SiMo_{12}O_{40}] + Ar_2I^+ \rightarrow \{H_4[SiMo_{12}O_{40}]\}^{\bullet+} + Ar_2I^\bullet\ (r2)$$

$$Ar_2I^\bullet \rightarrow Ar^\bullet + Ar\text{–}I\ (r3)$$

Parallel to this mechanism, POMs can also act as photosensitizer for the photoinduced decomposition of iodonium salts, as exemplified in equations r4 and r5 [37–40].

$$POM \rightarrow {}^*POM\ (h\nu)\ (r4)$$

$${}^*POM + Ar_2I^+ \rightarrow Ar^+ + Ar\text{-}I\ (r5)$$

In this field, an extensive work is done since several years by the group of Ortyl and coworkers to better sensitize iodonium salts in the visible range by developing appropriate photosensitizers [38,41–43].

In this work, the four POMs $[\alpha\text{-}PW_{11}VO_{40}]^{4-}$ ("$PW_{11}V$"), $[\alpha\text{-}PW_{11}Fe(H_2O)O_{39}]^{4-}$ ("$PW_{11}Fe$"), $[\gamma\text{-}SiW_{10}Fe_2(H_2O)_2O_{38}]^{6-}$ ("$SiW_{10}Fe_2$") and $[\gamma\text{-}SiW_{10}V_2O_{40}]^{6-}$ ("$SiW_{10}V_2$") were designed and synthesized as photoinitiators of polymerization in combination with an iodonium salt (see Fig. 1) either as counter cations (compounds denoted hereafter DPI-POMs) or in addition to tetrabutylammonium salts of POMs (denoted TBA-POMs).

Especially, their photoinitiating abilities were investigated as monocomponent or bi-component photoinitiating systems with the iodonium salt, considering that the solubility of POMs is a key-parameter governing the reactivity of the photoinitiating systems. Parallel to this, the different POMs were also used as photosensitizers for two benchmark photoinitiators, namely ethyl phenyl(2,4,6-trimethylbenzoyl) phosphinate (TPO-L) and 2-isopropyl-thioxanthone (2-ITX) (See Fig. 2). The free radical polymerization of a trifunctional acrylate monomer as well as the cationic polymerization of a difunctional epoxide monomer were examined upon excitation at 405 nm.

2. Experimental section

2.1. Materials

Trimethylolpropane triacrylate (TMPTA) was used as a trifunctional acrylate ester monomer and obtained from Sigma-Aldrich. 3,4-epoxycyclohexylmethyl 3,4-epoxycyclo-hexylcarboxylate (EPOX) was provided by Sartomer-Lambson UK. EPOX is a benchmark cycloaliphatic epoxy monomer. 2-Isopropylthioxanthone (2-ITX) was obtained from TCI-europe. Ethyl phenyl(2,4,6-trimethylbenzoyl)phosphinate (TPO-L) and the diphenyliodonium salt (Iod – Speedcure 938 also noted S938) was obtained from Lambson Ltd (see Fig. 2). POMs were prepared as tetrabutylammonium and diphenyliodonium salts and characterized by FT-IR, EDX, TGA, electrochemistry, elemental analyses (C,H,N), NMR (when possible) and UV–Visible spectroscopy. The protocols and the characterizations are given in the Supporting Information

2.2. Free radical photopolymerization experiments

For the different experiments, the photoinitiating system was dissolved in the acrylate monomer acrylate (TMPTA) and or in the acrylate/epoxy mixture (TMPTA/EPOX). The 1.4 mm thick samples were irradiated with a laser diode (LD) at 405 nm (110 $mW.cm^{-2}$). Irradiation at 405 nm was selected as a good example of mild and safe irradiation conditions (e.g. many consumer 3D printers are based on this wavelength). The evolution of the acrylate and epoxy function content was

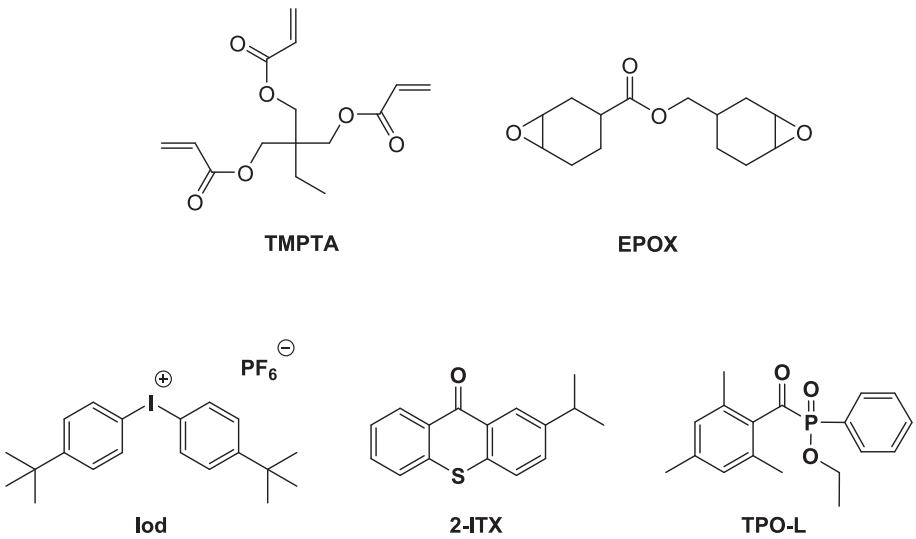

Fig. 2. Chemical structures of monomers and additives.

continuously monitored by real-time Fourier transform infrared spectroscopy (RT-FTIR, JASCO 4600 FTIR). The evolution of the acrylate peak was followed at [6117; 6223] cm^{-1} and [3700; 3790] cm^{-1} for epoxy. The procedure has been described in detail in [44–46].

3. Results and Discussion

3.1. Synthesis of the POM-iodonium salts and POM tetrabutylammonium salts

The POMs precursors $K_7[\alpha\text{-}PW_{11}O_{39}].14H_2O$ and $K_8[\gamma\text{-}SiW_{10}O_{36}].12H_2O$ were prepared by following the well-known protocols of the literature [47,48] and characterized by routine techniques (FT-IR, TGA, NMR) before using. The four POMs used in this study, *i.e.* $[\alpha\text{-}PW_{11}VO_{40}]^{4-}$, $[\alpha\text{-}PW_{11}Fe(H_2O)O_{39}]^{4-}$, $[\gamma\text{-}SiW_{10}Fe_2(H_2O)_2O_{38}]^{6-}$ and $[\gamma\text{-}SiW_{10}V_2O_{40}]^{6-}$, were prepared according to the literature procedures [49–52] by addition of aqueous solutions of $Fe(NO_3)_3$ or $NaVO_3$ to aqueous solutions of $K_7[\alpha\text{-}PW_{11}O_{39}].14H_2O$ or $K_8[\gamma\text{-}SiW_{10}O_{36}].12H_2O$ at the convenient pH value. The addition of tetrabutylammonium chloride or bromide or diphenyliodonium chloride in excess in water led to the precipitation of the tetrabutylammonium salts (noted TBA-POMs) or diphenyliodonium salts (noted DPI-POMs) of the POMs. After filtration and washing by ethanol and ether, the POM salts were characterized by FT-IR, EDX, TGA, elemental analyses (C,H,N) and NMR (when possible). The protocols and the characterizations are given in the Supporting Information

3.2. Light absorption properties

UV–visible absorption properties of the different DPI-POMs salts were investigated in acetonitrile. As shown in Fig. 3, all compounds exhibited a strong absorption in the UV range due to the W = O bonds and the π-π* transitions of the diphenyliodonium cation. Parallel to this, the three POMs $[Ph_2I]_4[\alpha\text{-}PW_{11}VO_{40}]$ (DPI-$PW_{11}V$), $[Ph_2I]_4H_2[\gamma\text{-}SiW_{10}Fe_2(H_2O)_2O_{38}]$ (DPI-$SiW_{10}Fe_2$), and $[Ph_2I]_{4.9}H_{1.1}[\gamma\text{-}SiW_{10}V_2O_{40}]$ (DPI-$SiW_{10}V_2$) also display an absorption tailing up 450 nm due to the charge transfer bands involving the Fe (III) or V (V) atoms. In the case of $[Ph_2I]_4[\alpha\text{-}PW_{11}Fe(H_2O)O_{39}]$ (DPI-$PW_{11}Fe$), only a weak absorption is detected in the visible range, not exceeding 425 nm. Based on their respective absorption, the different compounds could be examined as photoinitiators of polymerization at 405 nm. Furthermore, the gap energy values can be estimated by the Tauc's method [53]. These values are relatively high for compounds DPI-$PW_{11}V$ (3.6 eV) and DPI-$PW_{11}Fe$ (3.4 eV). The energy band gaps are lower for DPI-$SiW_{10}V_2$ (3.1 eV) and DPI-$SiW_{10}Fe_2$ (3.2 eV) and perfectly in line with values reported for TiO_2 (3.2 eV) and $(DPI)_4[W_{10}O_{32}]$ (3.3 eV) for which photochemical processes are reported in the literature [27,54].

3.3. Free radical polymerization at 405 nm using DPI – POMs

The four DPI-POMs, *i.e.* DPI-$PW_{11}V$, DPI-$PW_{11}Fe$, DPI-$SiW_{10}V_2$ and DPI-$SiW_{10}V_2$ were investigated as photoinitiators of polymerization for the free radical polymerization (FRP) of TMPTA in thick films under air, upon irradiation at 405 nm with a laser diode. A 3D polymer network is formed (noted poly(TMPTA)) with the formation of POM-poly(TMPTA) nanocomposites. Noticeably, despites the presence of the co-initiator as the counter-anion in these POMs, the different compounds could not initiate the FRP of TMPTA when used as monocomponent systems, attributable to the poor solubility of these structures in the acrylate monomer. Considering that the photoinduced electron transfer between POMs and the iodonium salt was inefficient, these structures were then tested as photosensitizers for two benchmark photoinitiators, namely 2-ITX and TPO-L. A summary of the TMPTA conversions is given in Table 1, using the two-component DPI-POM/2-ITX (1 %/2% w/w) and DPI-POM/TPO-L (1 %/2% w/w) systems. Noticeably, better TMPTA conversions could be obtained with the two-component DPI-POM/TPO-L (1 %/2% w/w) systems than the DPI-POM/2-ITX (1 %/2% w/w) systems. In all cases, a 2-fold enhancement of the monomer conversions was obtained using TPO-L as the photoinitiator. Thus, if a TMPTA conversion of only 40 % was obtained with the DPI-$PW_{11}Fe$/2-ITX system, this value increased to 93 % with the DPI-$PW_{11}Fe$/TPO-L system. The highest TMPTA conversion was obtained with DPI-$PW_{11}Fe$, peaking at 93 %. Interestingly, the two iron derivatives (DPI-$PW_{11}Fe$, DPI-$SiW_{10}Fe_2$) perform better than their vanadate analogues during the FRP of TMPTA. Comparison of the TMPTA conversions obtained with TPO-L, the reference TPO-L/Iod system and the DPI-POM/TPO-L systems revealed the contribution of POMs in the photoinitiation mechanism. As shown in Fig. 4 for DPI-$PW_{11}Fe$, a higher monomer conversion and a higher polymerization rate could be obtained with this POM.

3.4. Cationic polymerization at 405 nm using DPI – POMs

Abilities of the POM-iodonium salts, DPI-POMs, to initiate the cationic polymerization (CP) of epoxides was examined with a difunctional monomer, namely EPOX. All polymerization experiments were carried out at 405 nm. As previously observed during the FRP of TMPTA, none of the DPI-POM salts could initiate the CP of EPOX and assigned to the low solubility of these salts in EPOX, adversely affecting their

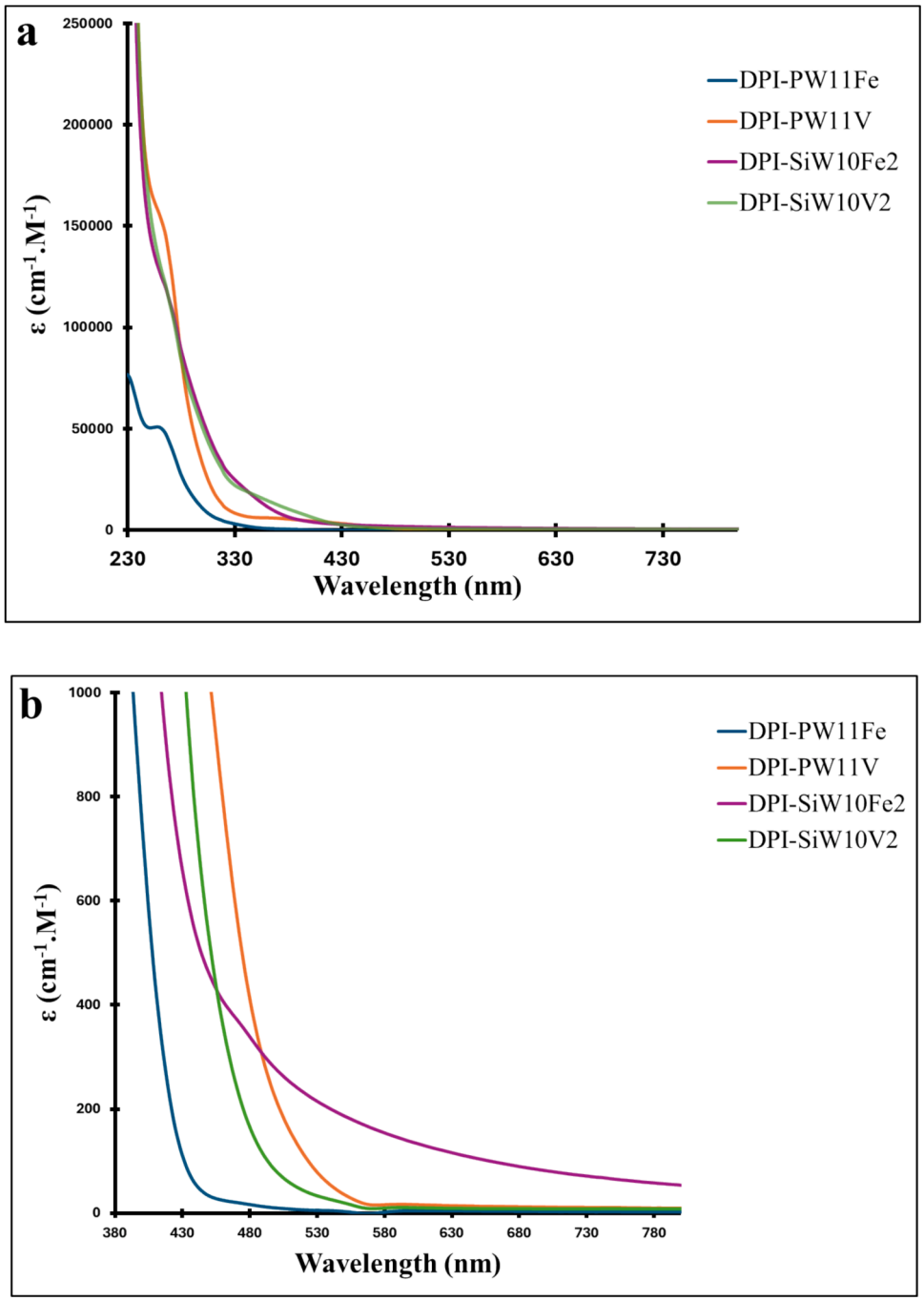

Fig. 3. UV–visible absorption spectra of the DPI-POM salts. **(a)** In DMSO at a concentration of 10^{-5} M. **(b)** In a DMSO/acetonitrile mixture (1/9 V/V) at a concentration of 10^{-3} M.

Table 1
Acrylate function conversions in TMPTA obtained with the two-component DPI-POM/2-ITX (1 %/2% w/w) and DPI-POM/TPO-L (1 %/2% w/w) systems upon excitation at 405 nm for 300 s.

Photosensitizers	2-ITX	TPO-L
DPI – $PW_{11}Fe$	40 %	93 %
DPI – $PW_{11}V$	55 %	87 %
DPI – $SiW_{10}Fe_2$	42 %	90 %
DPI – $SiW_{10}V_2$	40 %	86 %
Iod	–	83 %

photoinitiating abilities. When used in combination with 2-ITX, a different situation was found and a clear monomer conversion could be detected, using the two-component DPI-POM/2-ITX (1 %/2% w/w) system upon irradiation at 405 nm for 400 s in laminate (see Table 2). As anticipated, a severe reduction of the EPOX conversion was detected under air, evidencing that the EPOX conversion was resulting from the free radical promoted cationic polymerization (FRPCP) mechanism

During the CP of EPOX, all vanadate derivatives outperformed their iron analogues. Thus, if an EPOX conversion of 47 % was obtained with the DPI-$SiW_{10}Fe_2$/2-ITX system, this value increased up 86 % using the two-component DPI-$SiW_{10}V_2$/2-ITX system. Irrespective of the POM used, the DPI-POM/2-ITX systems furnished higher EPOX conversions than Iod (37 %), attributable to improved light absorption properties

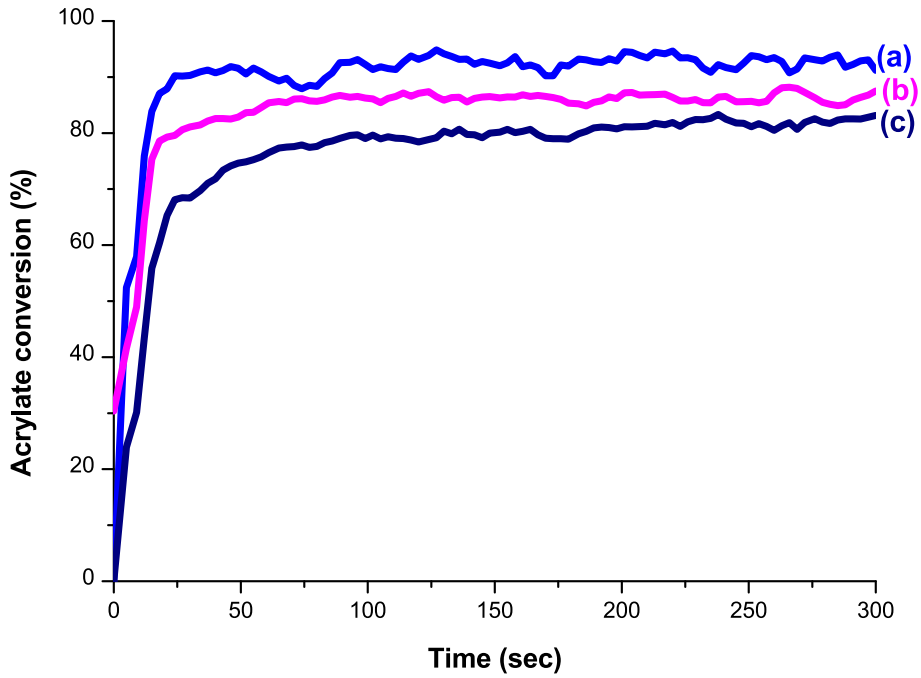

Fig. 4. Photopolymerization profiles (acrylate function conversion vs. irradiation time) (1.4 mm thick films, under air) upon exposure to a laser diode emitting at 405 nm (I = 110 mW.cm^{-2}) in TMPTA: (a) DPI-PW_{11} Fe/TPO-L (2 %/2% w/w); (b) TPO-L (2 wt%); (c) TPO-L/Iod (2 %/2% w/w). The irradiation starts at t = 10 s.

Table 2
EPOX conversions obtained with the two-component DPI-POM/2-ITX (1 %/2% w/w) systems upon excitation at 405 nm for 400 s.

Photosensitizers	EPOX conversion
DPI-PW_{11}Fe	47 %
DPI-PW_{11}V	67 %
DPI-$SiW_{10}Fe_2$	47 %
DPI-$SiW_{10}V_2$	86 %
Iod alone	37 %

Table 3
Acrylate function conversions in TMPTA obtained with the two-component DPI-POM/TPO-L (1 %/2% w/w), three-component TBA-POM/TPO-L/Iod (2 %/2%/2% w/w/w) and two-component TBA-POM/Iod (2 %/2% w/w) systems upon excitation at 405 nm for 300 s.

	DPI salts	TBA salts	
Photosensitizers	With TPO – L	With TPO – L	Without TPO – L
PW_{11}Fe	93 %	70 %	87 %
PW_{11}V	87 %	87 %	83 %
$SiW_{10}Fe_2$	90 %	92 %	87 %
$SiW_{10}V_2$	86 %	75 %	83 %

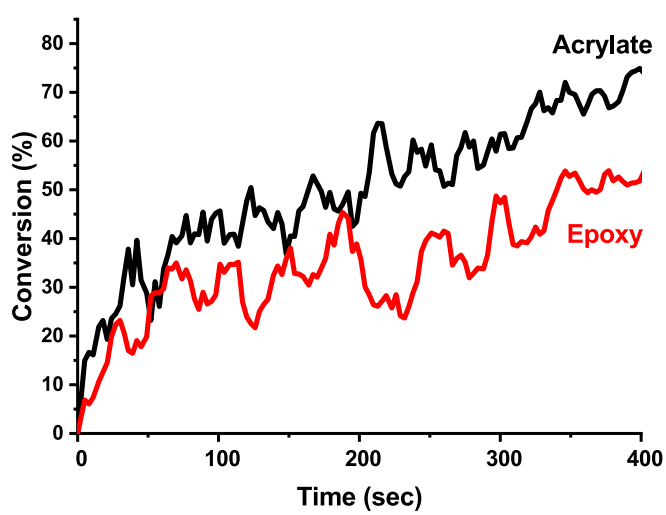

Fig. 5. Photopolymerization profiles (acrylate and epoxy function conversion vs. irradiation time) (1.4 mm thick films, under air) upon exposure to the LED@405 nm (I = 110 mW.cm^{-2}) in an EPOX/TMPTA blend using the two-component DPI-SiW_{10} V_2 /2-ITX (2 %/2% w/w) system. The irradiation starts at t = 10 s.

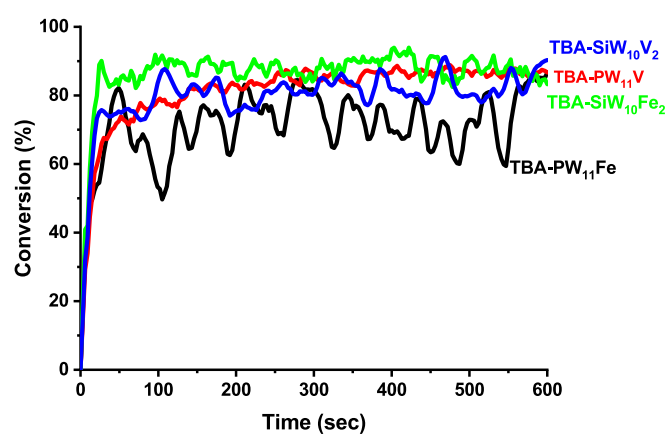

Fig. 6. TMPTA conversions determined upon excitation at 405 nm of thick films (1.4 mm) with a laser diode (110 mW/cm^2) using the three-component TBA-POM/TPO-L/Iod (2 %/2%/2% w/w/w) systems. The irradiation starts at t = 10 s.

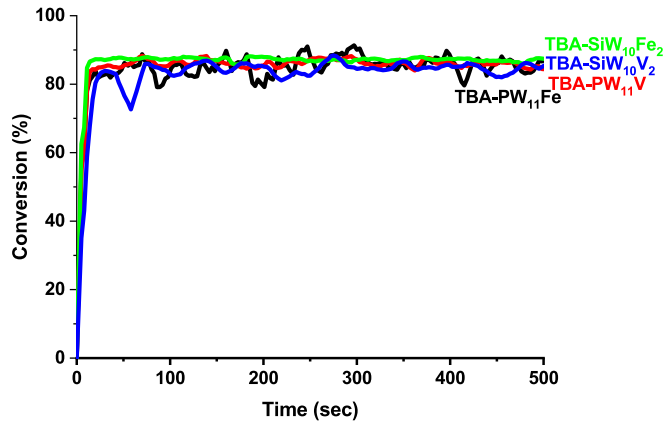

Fig. 7. Acrylate function conversions in TMPTA determined upon excitation at 405 nm of thick films (1.4 mm) with a laser diode (110 mW/cm^2) using the two-component TBA-POM/Iod (2 %/2% w/w) systems. The irradiation starts at t = 10 s.

resulting from the presence of POMs.

3.5. Formation of interpenetrating networks at 405 nm using DPI – POMs

Considering that the FRP of TMPTA and the CP of EPOX could be both promoted with the DPI-POM/2-ITX systems, the possibility of obtaining interpenetrating networks (IPN) by the concomitant polymerization of TMPTA and EPOX was studied. Here again, polymerization experiments were done at 405 nm. From a technical viewpoint, the photopolymerization of a mixture of multifunctional monomers results in the formation of cross-linked networks with specific properties [55–57]. The polymer obtained by the association of two types of monomers (radical and cationic here) benefits from the reunification of several properties within a single material. A 3D IPN is formed with the formation of POM-IPN nanocomposites. For the generation of IPNs, the best candidate was selected, namely DPI-$SiW_{10}V_2$.

As shown in Fig. 5, after 400 s of irradiation of thick films, 75 % conversion could be reached for the acrylate functions and approximately 55 % for the epoxy functions using the two-component DPI-$SiW_{10}V_2$/2-ITX (2 %/2% w/w) system.

Even if tack-free polymers could be obtained with the DPI-$SiW_{10}V_2$/2-ITX combination, the polymerization was slow, attributable to a poor solubility of DPI-$SiW_{10}V_2$ in resins. In order to investigate this point, a second series of POMs was synthesized, bearing tetrabutylammonium (TBA) cations, *i.e.* TBA-POM. These cations with long alkyl chains are well-known to improve the solubility of POMs in common organic solvents but also in monomers. Four TBA-POM salts were synthesized, namely TBA-$PW_{11}Fe$ ($[TBA]_4(\alpha–PW_{11}FeO_{40})$), TBA-$PW_{11}V$ ($[TBA]_4(\alpha–PW_{11}VO_{40})$), TBA-$SiW_{10}Fe_2$ ($[TBA]_{3.4}H_{2.6}[\gamma–SiW_{10}Fe_2O_2(H_2O)_2O_{36}]$) and TBA-$SiW_{10}V_2$ ($[TBA]_4H_2[\gamma–SiW_{10}V_2O_{40}]$).

3.6. Free radical polymerization at 405 nm using TBA – POMs/Iod

The FRP of TMPTA was carried out at 405 nm in thick films using the three-component TBA-POM/TPO-L/Iod (2 %/2%/2% w/w/w) systems, *i.e.* in conditions comparable to that used with the POM-iodonium salts (DPI-POMs) in section 3.3. Noticeably, a double bond conversion comparable to that obtained in Section 3.3 could be evidenced. The best monomer conversions were obtained with TBA-$SiW_{10}Fe_2$ and TBA-$PW_{11}V$, with conversions of 92 % for TBA-$SiW_{10}Fe_2$ (vs. 90 % with DPI-$SiW_{10}Fe_2$) and 87 % for TBA-$PW_{11}V$ (vs. 87 % with DPI-$PW_{11}V$) after 400 s of irradiation (See Table 3). A severe reduction of the TMPTA conversion could be detected for TBA-$PW_{11}Fe$, around 70 % conversion,

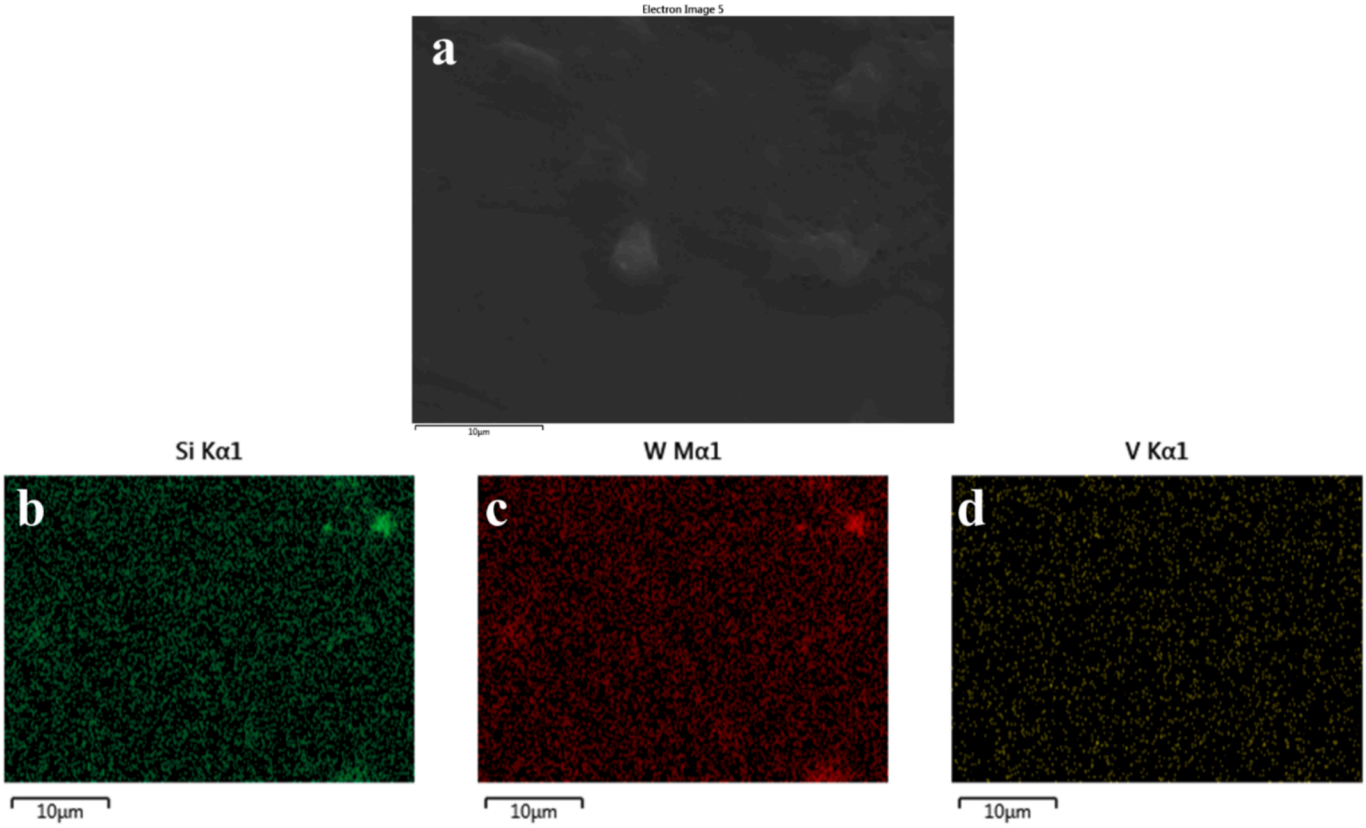

Fig. 8. SEM images of TBA – SiW_{10} V_2 – poly(TMPTA) composite at 10 µm **(a)**. EDX dot maps of elements Si **(b)**, W **(c)** and V **(d)** in TBA – SiW_{10} V_2 – poly (TMPTA) composite.

whereas a conversion of 93 % was obtained using the POM-iodonium salt DPI-$PW_{11}Fe$ (See Fig. 6).

Finally, the photoinitiating of the two-component TBA-POM/Iod system was examined. If no polymerization could be detected with the monocomponent DPI-POM salts without TPO-L, a different situation was found for the two-component TBA-POM /Iod systems. As shown in Fig. 7, all TBA-POM salts could efficiently promote the FRP of TMPTA, with a conversion of 83 % for TBA-$PW_{11}V$ and TBA-$SiW_{10}V_2$ up to 87 % for TBA-$PW_{11}Fe$ and TBA-$SiW_{10}Fe_2$. Especially, a high polymerization rate could be detected with the different two-component systems, enabling to reach the maximum conversion within 20 s.

The main difference between the monocomponent DPI-POM salts and the two-component TBA-POM/Iod systems being the solubility of the polyoxometalate salts in resins, the benefit of the tetrabutylammonium counter-cations in TBA-$SiW_{10}Fe_2$, TBA-$PW_{11}Fe$, TBA-$PW_{11}V$ and TBA-$SiW_{10}V_2$ was thus clearly demonstrated.

Fig. 8 shows SEM images of the TBA-$SiW_{10}V_2$ – poly(TMPTA) composite at scale of 10 μm, indicating that the polymer surface is generally smooth. Furthermore, a homogeneous dispersion of POMs within the polymer matrix is observed. Further images, along with TGA curves for both the TGA-POM-poly(TMPTA) composite and pure poly(TMPTA), are provided in the supporting information, confirming the successful incorporation of POMs within the polymer (Fig. S22).

4. Conclusion

A series of POM-iodonium salts has been designed and synthesized as visible light of polymerization. Investigation of their photoinitiating abilities in TMPTA and EPOX revealed these mono-component systems to be unable to initiate any polymerization due to solubility issues in resins. When combined with TPO-L, the different polyoxotungstates substituted by Iron(III) or Vanadium(V) metals could act as efficient photosensitizers for TPO-L, enabling to improve the TMPTA conversions compared to that obtained with the two-component TPO-L/Iod systems. To overcome the solubility issue of the monocomponent systems, a series of polyoxometalates bearing tetrabutylammonium cations was designed and synthesized. Due to an improved solubility in resins, the different POM-TBA salts could promote the FRP of TMPTA in two-component TBA-POM/Iod systems, whereas no polymerization was detected with the monocomponent POM-iodonium salts. Despite a significant improvement of the solubility by use of tetrabutylammonium salts, the solubility of the different polyoxotungstates substituted by Iron (III) or Vanadium (V) metals was still far behind that of the previous POMs reported in the literature, adversely affecting their photoinitiating abilities. Based on the present results obtained with these structures, excellent monomer conversions could be obtained in the Future by improving the solubility of polyoxotungstates substituted by Iron (III) or Vanadium (V) metals in resins.

CRediT authorship contribution statement

Roa AlChamandi: Formal analysis, Conceptualization. **Tuba Kavalli:** Conceptualization. **Corine Simonnet-Jégat:** Data curation. **Nathalie Leclerc:** Formal analysis, Conceptualization. **Catherine Roch-Marchal:** Formal analysis. **Frédéric Dumur:** Data curation. **Akram Hijazi:** Data curation. **Jacques Lalevée:** Data curation, Conceptualization. **Sébastien Floquet:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2025.113713.

Data availability

Data will be made available on request.

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